16S LiFePO4 BMS - Total System Bill of Materials (Encompassing Everything)

Master Procurement Schedule across all 8 Schematic Subsystems (Cell AFE, Power Supply, MCU, Switching, Temp Sense, Balance LEDs, Isolated CAN/Comms, Precharge) & External Hardware

Document Ref: 16S-BMS-TOTAL-BOM
Date: November 2026 | Rev: 3.0 (Full 100% Audit)
Scope: Complete System (All Sources)

ET Store	Evelta	Robu.in	LionCircuits	Rajiv Electronics	Sharvi	TOTAL SYSTEM
etstore.in	evelta.com	robu.in	lioncircuits.com	rajivelectronics.com	sharvielectronics.com	6 Indian Sources
103 Lines | 280 pcs	7 Lines | 7 pcs	5 Lines | 6 pcs	1 Line | 1 pc	7 Lines | 19 pk	1 Line | 1 pc	124 Lines | 302 pcs
₹ 3,407.92	₹ 814.64	₹ 775.00	₹ 665.29	₹ 448.10	₹ 184.88	₹ 6,295.83*

#	Subsystem	Category	Designator(s)	Component / Val	Description & Purpose	Package / Footprint	Mnt	D. Qty	O. Qty	Unit (₹)	Total (₹)	Supplier
1	External (Chassis / Busbar)	Current Shunt	RS1 (External Current Shunt, wired to J31/J32)	FL-2B 100A 75mV	XIQI Electric CG FL-2B 100A 75mV DC Current Shunt Resistor	Chassis_BoltMount	Chassis/ THT	1	1	352.00	352.00	**robu.in**
2	External (Chassis / Precharge)	Precharge Resistor	R_PRE (External Precharge Chassis-Mount Power Resistor, wired to J35)	50 Ω 25W	50 Ohm 25W Aluminum Housed Wirewound Power Resistor (55mm mounting pitch)	Chassis_Mount_55mm_Pitch	Chassis/ THT	1	1	184.88	184.88	**sharvielectronics.com**
3	External (Cell Pack Harness)	Sensors (External)	TH1..TH16 (16x Cell Temperature Sensors for DB-25 J20)	10k NTC 3950 (with wire)	10K Ohm NTC Thermistor Temperature Sensor Probe with 1m wire lead	Probe_Wire_Harness	Harness	16	16	14.58	233.28	etstore.in
4	01 Cell AFE	ICs & Modules	U1	BQ76952PFBR	16S LiFePO4 Battery Monitor, Protector & Balancer IC	QFP50P900X900X120-48N	SMD	1	1	665.29	665.29	**lioncircuits.com**
5	04 Current Switching	ICs & Modules	U2, U3	UCC27517DBVR	4A High-Speed Low-Side Gate Driver, SOT-23-5	SOT-23-5	SMD	2	2	35.00	70.00	**robu.in**
6	04 Current Switching	ICs & Modules	U4, U5, U6, U7, U40	EL817	Single Channel Phototransistor Optocoupler, SOP-4	SOP-4_3.8x4.1mm_P2.54mm	SMD	5	6	6.50	39.00	etstore.in
7	05 Temperature Sensing	ICs & Modules	U8, U9, U30	CD74HC4067SM	High-Speed 16-Channel Analog Multiplexer / Demultiplexer	SSOP-24_5.3x8.2mm_P0.65mm	SMD	3	3	56.87	170.61	etstore.in
8	02 Power Supply	ICs & Modules	U10	LM5164DDA	100V 1A Synchronous Step-Down Buck DC-DC Converter	HSOP-8-1EP_3.9x4.9mm_P1.27mm_EP2.41x3.1mm_ThermalVias	SMD	1	1	548.70	548.70	**evelta.com**
9	02 Power Supply	ICs & Modules	U11	LM2596S-3.3	3A Step-Down Voltage Regulator (LM2596S-3.3), TO-263-5	TO-263-5_TabPin3	SMD	1	1	35.00	35.00	etstore.in
10	02 Power Supply	ICs & Modules	U12	B1212S-1WR3	1W 12V-to-12V Isolated DC-DC Converter Module	Converter_DCDC_Murata_MEE1Sxxx xSC_THT	THT	1	1	103.00	103.00	etstore.in
11	02 Power Supply	ICs & Modules	U13	LM2596S-5.0	3A Step-Down Voltage Regulator (LM2596S-5.0), TO-263-5	TO-263-5_TabPin3	SMD	1	1	35.00	35.00	etstore.in
12	07 Isolated CAN & Comms	ICs & Modules	U13	ISO1050DUBR	5V Isolated CAN Transceiver, SOP-8	SOP-8_6.62x9.15mm_P2.54mm	SMD	1	1	230.10	230.10	**evelta.com**
13	07 Isolated CAN & Comms	ICs & Modules	U14	B1209S-1WR3	1W 12V-to-9V Isolated DC-DC Converter for Gate Drive Rail	Converter_DCDC_Murata_MEE1Sxxx xSC_THT	THT	1	1	109.00	109.00	**robu.in**
14	07 Isolated CAN & Comms	ICs & Modules	U15	AMS1117-5.0	1A Low Dropout Linear Voltage Regulator 5.0V, SOT-223	SOT-223-3_TabPin2	SMD	1	1	8.00	8.00	etstore.in
15	03 MCU Control	ICs & Modules	U20	CH32V203C8T6	QingKe 32-bit RISC-V MCU, 64KB Flash, 20KB SRAM, LQFP-48	LQFP-48_7x7mm_P0.5mm	SMD	1	1	95.00	95.00	etstore.in
16	03 MCU Control	ICs & Modules	U21	FM24CL64B-G	64 Kbit (8Kx8) High-Endurance Non-Volatile I2C FRAM	SOIC-8_3.9x4.9mm_P1.27mm	SMD	1	1	232.00	232.00	**robu.in**
17	06 Balance Status	ICs & Modules	U31..U34 (16 items: U31, U31, U31, U31 ...)	LTV-847S	Lite-On 4-Channel DC Optocoupler Isolator (Quad)	SMDIP-16_W9.53mm	SMD	4	4	60.53	242.12	etstore.in
18	04 Current Switching	Transistors/FETs	Q1..Q19 (17 items: Q1, Q2, Q3, Q4 ...)	IRFB4110	100V 180A 3.7mR N-Channel Power MOSFET (TO-220 THT)	TO-220-3_Vertical	THT	17	17	55.99	951.83	etstore.in
19	04 Current Switching	Transistors/FETs	Q17, Q18	BC817-40,215	45V 500mA General Purpose NPN Transistor, SOT-23	SOT-23_Handsoldering	SMD	2	5	3.00	15.00	etstore.in
20	01 Cell AFE	Transistors/FETs	Q21	BCX56-135	80V 1A Medium Power NPN Transistor, SOT-89-3	SOT-89-3_Handsoldering	SMD	1	1	11.38	11.38	**evelta.com**

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21	01 Cell AFE	Diodes/TVS	D1, D2	BAV21WS	250V 200mA Fast Switching Diode, SOD-323	D_SOD-323_HandSoldering	SMD	2	5	2.50	12.50	etstore.in
22	02 Power Supply	Diodes/TVS	D3	SMBJ60CA	60V 600W Bidirectional TVS Diode, SMB	D_SMB	SMD	1	1	15.00	15.00	**evelta.com**
23	02 Power Supply	Diodes/TVS	D4	B340	40V 3A Schottky Barrier Rectifier Diode, SMC	D_SMC	SMD	1	2	6.00	12.00	etstore.in
24	02 Power Supply	Diodes/TVS	D5	B340	40V 3A Schottky Barrier Rectifier Diode, SMC	D_SMC	SMD	1	2	6.00	12.00	etstore.in
25	07 Isolated CAN & Comms	Diodes/TVS	D5	PESD2CAN	CAN Bus Dual-line ESD Protection TVS Diode, SOT-23	SOT-23	SMD	1	1	12.00	12.00	**robu.in**
26	02 Power Supply	Diodes/TVS	D6	B5819W	40V 1A Surface Mount Schottky Barrier Diode, SOD-123	D_SMA	SMD	1	5	3.00	15.00	etstore.in
27	03 MCU Control	Diodes/TVS	D10	GREEN	SMD Green Indicator LED, 0805	LED_0805_2012Metric	SMD	1	5	2.00	10.00	etstore.in
28	07 Isolated CAN & Comms	Diodes/TVS	D20	SMAJ6.0A	6.0V 400W Fast Unidirectional TVS Diode, SMA	D_SMA	SMD	1	1	4.15	4.15	**evelta.com**
29	08 Precharge Circuit	Diodes/TVS	D40	SMAJ75A	75V 400W Fast Unidirectional TVS Diode (Precharge Inductive Clamp, SMA)	D_SMA	SMD	1	1	3.30	3.30	**evelta.com**
30	08 Precharge Circuit	Diodes/TVS	D41	BZT52C15	15V 500mW SMD Zener Diode (Gate Clamp, SOD-123)	D_SOD-123	SMD	1	5	1.77	8.85	etstore.in
31	08 Precharge Circuit	Diodes/TVS	D42	BAT54S	Dual Schottky Barrier Diode Series Pair (Precharge Sense ADC Rail Clamp, SOT-23)	SOT-23	SMD	1	1	2.01	2.01	**evelta.com**
32	02 Power Supply	Protection/Fuses	F1	2A 80V	2A 80V Fast-Acting Surface Mount Fuse, 1812	Fuse_1812_4532Metric	SMD	1	2	16.00	32.00	etstore.in
33	02 Power Supply	Protection/Fuses	F2	3.15A	3.15A Fast-Acting Surface Mount Fuse, 1812	Fuse_1812_4532Metric	SMD	1	2	16.00	32.00	etstore.in
34	07 Isolated CAN & Comms	Protection/Fuses	F10	1.0A 1812 PTC	1.0A 16V Resettable Polymeric PTC Fuse, 1812	Fuse_1812_4532Metric	SMD	1	2	18.00	36.00	etstore.in
35	External (Harness Cable)	Connectors/Hardware	P20 (Mating Plug for J20 DB-25 NTC Harness)	DB25 Male Solder Plug + Hood	DB25 Male Solder-Type Connector Plug with Plastic/Metal Protective Shell	Cable_Mount	Cable	1	1	35.00	35.00	**rajivelectronics.com**
36	External (Harness Cable)	Connectors/Hardware	P30 (Mating Plug for J30 DD-50 Balance Tap Harness)	DB50 Male Solder Plug + Hood	DB50 50-Pin D-Sub Male Solder-Type Connector Plug with Protective Dust Cover	Cable_Mount	Cable	1	1	95.00	95.00	**rajivelectronics.com**
37	02 Power Supply	Connectors/Hardware	J1	FAN1	2-Pin 5.08mm Pitch PCB Screw Terminal Block	TerminalBlock_Phoenix_MKDS-1,5-2-5.08_1x02_P5.08mm_Horizontal	THT	1	1	12.00	12.00	etstore.in
38	02 Power Supply	Connectors/Hardware	J2	FAN2	2-Pin 5.08mm Pitch PCB Screw Terminal Block	TerminalBlock_Phoenix_MKDS-1,5-2-5.08_1x02_P5.08mm_Horizontal	THT	1	1	12.00	12.00	etstore.in
39	02 Power Supply	Connectors/Hardware	J4	12V_EXT	2-Pin 5.08mm Pitch PCB Screw Terminal Block	TerminalBlock_Phoenix_MKDS-1,5-2-5.08_1x02_P5.08mm_Horizontal	THT	1	1	12.00	12.00	etstore.in
40	07 Isolated CAN & Comms	Connectors/Hardware	J4	PACK LINK, NOT ETHERNET	RJ45 8P8C Right-Angle Shielded PCB Modular Jack	RJ45_Amphenol_54602-x08_Horizontal	THT	1	1	25.00	25.00	etstore.in
41	03 MCU Control	Connectors/Hardware	J10	WCH-LinkE	2.54mm Pitch Breakaway Pin Header (WCH-LinkE)	PinHeader_1x05_P2.54mm_Vertical	THT	1	1	5.00	5.00	etstore.in
42	03 MCU Control	Connectors/Hardware	J11	Debug UART	2.54mm Pitch Breakaway Pin Header (Debug UART)	PinHeader_1x03_P2.54mm_Vertical	THT	1	1	5.00	5.00	etstore.in
43	03 MCU Control	Connectors/Hardware	J12	OLED Display (I2C)	0.96 inch 128x64 I2C OLED Display Module (SSD1306, 4-Pin)	PinHeader_1x04_P2.54mm_Vertical	THT	1	1	171.73	171.73	etstore.in
44	06 Balance Status	Connectors/Hardware	J20	DB25_Female	25-Pin D-Sub Female Right-Angle PCB Connector (NTC Taps)	DSUB-25_Socket_Horizontal_P2.77x2.84mm_EdgePinOffset4.94mm_Housed_MountingHolesOffset4.94mm	THT	1	1	26.00	26.00	**rajivelectronics.com**
45	Top Level Interconnect	Connectors/Hardware	J30	DD50_Female	50-Pin D-Sub High-Density Female PCB Connector (Cell Taps)		THT	1	1	112.10	112.10	**rajivelectronics.com**
46	Top Level Interconnect	Connectors/Hardware	J31	SHUNT KELVIN	2.54mm Pitch Breakaway Pin Header (SHUNT KELVIN)	PinHeader_1x02_P2.54mm_Vertical	THT	1	1	5.00	5.00	etstore.in
47	Top Level Interconnect	Connectors/Hardware	J32	FROM SHUNT	FROM SHUNT High-Current M5/M6 Direct Bolt-Down PCB Pad Terminal	MountingHole_8.4mm_M8_Pad	THT/Pad	1	5	12.00	60.00	**rajivelectronics.com**

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48	Top Level Interconnect	Connectors/Hardware	J33	PACK MINUS	PACK MINUS High-Current M5/M6 Direct Bolt-Down PCB Pad Terminal	MountingHole_8.4mm_M8_Pad	THT/Pad	1	5	12.00	60.00	**rajivelectronics.com**
49	Top Level Interconnect	Connectors/Hardware	J34	PACK PLUS TAP	PACK PLUS TAP High-Current M5/M6 Direct Bolt-Down PCB Pad Terminal	MountingHole_8.4mm_M8_Pad	THT/Pad	1	5	12.00	60.00	**rajivelectronics.com**
50	08 Precharge Circuit	Connectors/Hardware	J35	AK300-2	2-Pin 5.08mm Pitch Precharge Resistor Terminal Block (Chassis Resistor Header)	TerminalBlock_Phoenix_MKDS-1,5-2-5.08_1x02_P5.08mm_Horizontal	THT	1	1	12.00	12.00	etstore.in
51	02 Power Supply	Connectors/Hardware	JP1	JP_OPEN	2-Pin Solder Jumper (JP_OPEN)	SolderJumper-2_P1.3mm_Open_Pad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
52	07 Isolated CAN & Comms	Connectors/Hardware	JP1	TERMINATION	2-Pin Solder Jumper (TERMINATION)	SolderJumper-2_P1.3mm_Open_RoundedPad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
53	03 MCU Control	Connectors/Hardware	JP10	BOOT0	2-Pin Solder Jumper (BOOT0)	SolderJumper-2_P1.3mm_Open_RoundedPad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
54	03 MCU Control	Connectors/Hardware	JP11	ADDR0	2-Pin Solder Jumper (ADDR0)	SolderJumper-2_P1.3mm_Open_RoundedPad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
55	03 MCU Control	Connectors/Hardware	JP12	ADDR1	2-Pin Solder Jumper (ADDR1)	SolderJumper-2_P1.3mm_Open_RoundedPad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
56	03 MCU Control	Connectors/Hardware	JP13	ADDR2	2-Pin Solder Jumper (ADDR2)	SolderJumper-2_P1.3mm_Open_RoundedPad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
57	03 MCU Control	Connectors/Hardware	JP14	EE_WP	2-Pin Solder Jumper (EE_WP)	SolderJumper-2_P1.3mm_Open_RoundedPad1.0x1.5mm	SMD	1	0	0.00	0.00	etstore.in
58	03 MCU Control	Inductors/Filters	FB10	120R/1.5A	120 Ohm @ 100MHz 1.5A Ferrite Bead, 0805	L_0805_2012Metric	SMD	1	5	3.50	17.50	etstore.in
59	02 Power Supply	Inductors/Filters	L1	68 uH	68 uH High-Current Shielded Power Inductor (SRR1260)	L_Bourns_SRR1260	SMD	1	1	28.00	28.00	etstore.in
60	02 Power Supply	Inductors/Filters	L2, L3	33 uH	33 uH High-Current Shielded Power Inductor (SRR1260)	L_Bourns_SRR1260	SMD	2	2	28.00	56.00	etstore.in
61	03 MCU Control	Crystals	Y1	8MHz	8.000 MHz Fundamental SMD Crystal Resonator (3225-4P)	Crystal_SMD_3225-4Pin_3.2x2.5mm	SMD	1	1	15.00	15.00	etstore.in
62	01 Cell AFE	Resistors	R1..R17 (17 items: R1, R2, R3, R4 ...)	20 Ω	SMD Chip Resistor 20 Ω, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	17	20	0.75	15.00	etstore.in
63	01 Cell AFE	Resistors	R18, R21, R22	100 Ω	SMD Chip Resistor 100 Ω, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	3	10	0.75	7.50	etstore.in
64	01 Cell AFE	Resistors	R19..R58 (22 items: R19, R20, R23, R24 ...)	10 kΩ	SMD Chip Resistor 10 kΩ, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	22	30	0.75	22.50	etstore.in
65	04 Current Switching	Resistors	R25..R40 (16 items: R25, R26, R27, R28 ...)	27 Ω	SMD Chip Resistor 27 Ω, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	16	20	0.75	15.00	etstore.in
66	04 Current Switching	Resistors	R59, R61	2.2 kΩ	SMD Chip Resistor 2.2 kΩ, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	2	10	0.75	7.50	etstore.in
67	05 Temperature Sensing (NTC Dividers)	Resistors	R62..R110 (32 items: R62, R64, R66, R73 ...)	10 kΩ	SMD Chip Resistor 10 kΩ, 1% tolerance	R_0805_2012Metric_Pad1.20x1.40mm_HandSolder	SMD	32	40	0.75	30.00	etstore.in
68	04 Current Switching	Resistors	R63, R65	100 kΩ	SMD Chip Resistor 100 kΩ, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	2	10	0.75	7.50	etstore.in
69	04 Current Switching	Resistors	R67, R68, R71, R72	330 Ω	SMD Chip Resistor 330 Ω, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	4	10	0.75	7.50	etstore.in
70	01 Cell AFE	Resistors	R69	300 Ω	SMD Chip Resistor 300 Ω, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	1	10	0.75	7.50	etstore.in
71	01 Cell AFE	Resistors	R70	10 MΩ	SMD Chip Resistor 10 MΩ, 1% tolerance	R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	SMD	1	10	0.75	7.50	etstore.in
72	08 Precharge Circuit	Resistors	R90	220	SMD Chip Resistor 220, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in
73	08 Precharge Circuit	Resistors	R91	100k	SMD Chip Resistor 100k, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in

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74	08 Precharge Circuit	Resistors	R92	100	SMD Chip Resistor 100, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in
75	08 Precharge Circuit	Resistors	R93	10k	SMD Chip Resistor 10k, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in
76	08 Precharge Circuit	Resistors	R94, R96	84.5k 1%	SMD Chip Resistor 84.5k 1%, 1% tolerance	R_1206_3216Metric	SMD	2	10	0.75	7.50	etstore.in
77	08 Precharge Circuit	Resistors	R95	10k 1%	SMD Chip Resistor 10k 1%, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in
78	08 Precharge Circuit	Resistors	R97	1k	SMD Chip Resistor 1k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
79	02 Power Supply	Resistors	R101	1M	SMD Chip Resistor 1M, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in
80	02 Power Supply	Resistors	R102	51k	SMD Chip Resistor 51k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
81	02 Power Supply	Resistors	R103, R121, R126, R127	100k	SMD Chip Resistor 100k, 1% tolerance	R_0805_2012Metric	SMD	4	10	0.75	7.50	etstore.in
82	02 Power Supply	Resistors	R104	270k	SMD Chip Resistor 270k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
83	02 Power Supply	Resistors	R105	30k	SMD Chip Resistor 30k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
84	02 Power Supply	Resistors	R106	470k	SMD Chip Resistor 470k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
85	07 Isolated CAN & Comms	Resistors	R107, R120, R131	10k	SMD Chip Resistor 10k, 1% tolerance	R_0805_2012Metric	SMD	3	10	0.75	7.50	etstore.in
86	07 Isolated CAN & Comms	Resistors	R108	2k	SMD Chip Resistor 2k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
87	07 Isolated CAN & Comms	Resistors	R109	120	SMD Chip Resistor 120, 1% tolerance	R_1206_3216Metric	SMD	1	10	0.75	7.50	etstore.in
88	03 MCU Control	Resistors	R125	1k	SMD Chip Resistor 1k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
89	03 MCU Control	Resistors	R128, R129	4.7k	SMD Chip Resistor 4.7k, 1% tolerance	R_0805_2012Metric	SMD	2	10	0.75	7.50	etstore.in
90	06 Balance Status	Resistors	R130	47k	SMD Chip Resistor 47k, 1% tolerance	R_0805_2012Metric	SMD	1	10	0.75	7.50	etstore.in
91	01 Cell AFE	Capacitors	C1..C17 (17 items: C1, C2, C3, C4 ...)	220 nF	220 nF 50V Ceramic Decoupling Capacitor, 0805	C_1206_3216Metric_Pad1.33x1.80mm_HandSolder	SMD	17	20	1.25	25.00	etstore.in
92	01 Cell AFE	Capacitors	C18, C20	1 µF	1 µF 50V Ceramic Decoupling Capacitor, 0805	C_1812_4532Metric_Pad1.57x3.40mm_HandSolder	SMD	2	10	1.25	12.50	etstore.in
93	01 Cell AFE	Capacitors	C19	100 pF	100 pF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric_Pad1.18x1.45mm_HandSolder	SMD	1	10	1.25	12.50	etstore.in
94	01 Cell AFE	Capacitors	C21	2.2 µF	2.2 µF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric_Pad1.18x1.45mm_HandSolder	SMD	1	10	1.25	12.50	etstore.in
95	01 Cell AFE	Capacitors	C22	100 nF	100 nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric_Pad1.18x1.45mm_HandSolder	SMD	1	10	1.25	12.50	etstore.in
96	01 Cell AFE	Capacitors	C23	22 nF	22 nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric_Pad1.18x1.45mm_HandSolder	SMD	1	10	1.25	12.50	etstore.in
97	01 Cell AFE	Capacitors	C24	1 µF	1 µF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric_Pad1.18x1.45mm_HandSolder	SMD	1	10	1.25	12.50	etstore.in
98	05 Temperature Sensing	Capacitors	C25, C26	100 nF	100 nF 50V Ceramic Decoupling Capacitor, 0805	C_1206_3216Metric_Pad1.33x1.80mm_HandSolder	SMD	2	10	1.25	12.50	etstore.in
99	02 Power Supply	Capacitors	C27	4.7uF 100V	4.7uF 100V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_1210_3225Metric	SMD	1	10	3.50	35.00	etstore.in
100	02 Power Supply	Capacitors	C28	100nF 100V	100nF 100V 100V High-Voltage Ceramic Capacitor	C_0805_2012Metric	SMD	1	5	4.50	22.50	etstore.in
101	02 Power Supply	Capacitors	C29	2.2nF 50V	2.2nF 50V 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
102	02 Power Supply	Capacitors	C30	3.3nF	3.3nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
103	02 Power Supply	Capacitors	C31	56pF	56pF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
104	02 Power Supply	Capacitors	C32	22uF 25V	22uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_1206_3216Metric	SMD	1	10	3.50	35.00	etstore.in
105	02 Power Supply	Capacitors	C33, C35	100nF 25V	100nF 25V 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	2	10	1.25	12.50	etstore.in
106	02 Power Supply	Capacitors	C34	470uF 25V	470uF 25V Radial Aluminum Electrolytic Capacitor (Low-ESR)	CP_Elec_8x10	THT	1	2	8.50	17.00	etstore.in
107	02 Power Supply	Capacitors	C36	220uF 10V	220uF 10V Radial Aluminum Electrolytic Capacitor (Low-ESR)	CP_Elec_8x10	THT	1	2	8.50	17.00	etstore.in

#	Subsystem	Category	Designator(s)	Component / Val	Description & Purpose	Package / Footprint	Mnt	D. Qty	O. Qty	Unit (₹)	Total (₹)	Supplier
108	02 Power Supply	Capacitors	C37	100nF 10V	100nF 10V 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
109	02 Power Supply	Capacitors	C38, C39	4.7uF 25V	4.7uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_0805_2012Metric	SMD	2	10	3.50	35.00	etstore.in
110	02 Power Supply	Capacitors	C40	22uF 25V	22uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_1206_3216Metric	SMD	1	10	3.50	35.00	etstore.in
111	07 Isolated CAN & Comms	Capacitors	C40, C41	100nF	100nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	2	10	1.25	12.50	etstore.in
112	02 Power Supply	Capacitors	C41	220uF 10V	220uF 10V Radial Aluminum Electrolytic Capacitor (Low-ESR)	CP_Elec_8x10	THT	1	2	8.50	17.00	etstore.in
113	02 Power Supply	Capacitors	C42	100nF	100nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
114	07 Isolated CAN & Comms	Capacitors	C42	4.7uF 25V	4.7uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_0805_2012Metric	SMD	1	10	3.50	35.00	etstore.in
115	07 Isolated CAN & Comms	Capacitors	C43..C71 (12 items: C43, C45, C47, C50 ...)	100nF	100nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	12	20	1.25	25.00	etstore.in
116	07 Isolated CAN & Comms	Capacitors	C44	10uF 25V	10uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_1206_3216Metric	SMD	1	10	3.50	35.00	etstore.in
117	07 Isolated CAN & Comms	Capacitors	C46	22uF 10V	22uF 10V Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_1206_3216Metric	SMD	1	10	3.50	35.00	etstore.in
118	03 MCU Control	Capacitors	C53	4.7uF	4.7uF Multi-layer Ceramic Capacitor (MLCC), 1206/0805	C_0805_2012Metric	SMD	1	10	3.50	35.00	etstore.in
119	03 MCU Control	Capacitors	C55	1uF	1uF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
120	03 MCU Control	Capacitors	C56	10nF	10nF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	1	10	1.25	12.50	etstore.in
121	03 MCU Control	Capacitors	C57, C58	18pF	18pF 50V Ceramic Decoupling Capacitor, 0805	C_0805_2012Metric	SMD	2	10	1.25	12.50	etstore.in
122	04 Current Switching	Capacitors	C_DRV1, C_DRV4	100 nF	100 nF 50V Ceramic Decoupling Capacitor, 0805		SMD	2	10	1.25	12.50	etstore.in
123	04 Current Switching	Capacitors	C_DRV2, C_DRV3	4.7 µF	4.7 µF Multi-layer Ceramic Capacitor (MLCC), 1206/0805		SMD	2	10	3.50	35.00	etstore.in
124	02 Power Supply	PCB Layout	NT1, NT2, NT3	NetTie_2	Net Tie Trace Link (Copper feature on PCB)	NetTie-2_SMD_Pad0.5mm	SMD	3	0	0.00	0.00	etstore.in
TOTAL	**16S BMS Complete System (All 8 Sheets + External Harness & Hardware)**							302	819		**₹ 6,295.83***	**6 Vendors**

Procurement Guidelines & Commercial Packaging Notes:
1. Strict Supplier Boundary: Every component is exclusively sourced from user-approved domestic Indian distributors: ET Store (etstore.in), Evelta (evelta.com), Robu.in (robu.in), LionCircuits (lioncircuits.com), Rajiv Electronics (rajivelectronics.com), and Sharvi Electronics (sharvielectronics.com). International distributors (LCSC, Mouser, DigiKey) are excluded.
2. Packaging & MOQ Rule: Resistors and general capacitors are procured in commercial minimum packs of 10 units (or 5 units for power/electrolytic/TVS/fuse components). Terminal bolts (J32, J33, J34) are ordered in 5-unit hardware packs. Design Quantity = 302 pcs; Total Order Quantity = 819 pcs.
3. Precharge Circuit (Sheet 08): Precharge switch Q19 (IRFB4110), optocoupler U40 (EL817), zener D41 (BZT52C15), header J35 (5.08mm) and divider resistors (R90-R97) are on ET Store; TVS D40 (SMAJ75A) and ADC clamp D42 (BAT54S) are on Evelta; external power resistor R_PRE (47 ohm 25W/50W chassis mount) is on Sharvi Electronics.
4. Complete Coverage: All 8 KiCad schematic sheets, 32 NTC divider channels, off-board XIQI 100A current shunt, external 16-channel thermistor harness, and D-Sub mating hoods are 100% encompassed.